

Title

Comparative Evaluation of Intelligent Traffic Signal Control for Emergency Vehicle Prioritization with Blue Light indicators Using SUMO and TraCI

Introduction

Background

Traffic congestion and delays at intersections are significant concerns in urban transportation, especially when emergency vehicles (EVs) such as ambulances and fire trucks require uninterrupted and prioritized passage. Intelligent Traffic Signal Control (ITSC) systems can play a pivotal role in addressing these issues by adapting signal timing based on real-time traffic conditions.

This project presents a concept to extend existing intelligent traffic signal control algorithms such as Round Robin (RR), Feedback Control (MONOPOLY) and Advantage Actor-Critic (A2C). We aim to apply and evaluate this approach at a real-world intersection near ABS in Kronach, Germany, using simulation-based modeling using SUMO and TraCI.

Problem Statement

While several signal control strategies exist, there is limited comparative evaluation of their performance specifically for emergency vehicles operating with blue lights. Blue lights typically indicate a higher urgency, potentially altering vehicle behavior and traffic rules in real-world settings. The effectiveness of these control strategies must be measured through robust simulation tools to ensure efficient and safe EV prioritization.

Literature Review

Recent studies have demonstrated the potential of intelligent control systems to improve emergency vehicle (EV) prioritization. While our work focuses on a single-intersection evaluation using SUMO and TraCI, the following studies guide our approach:

1. VLMLight (Chen et al., 2024) proposed a meta-learning model for EV signal control using visual and language inputs. Although their setup is more complex, the core idea of prioritizing EVs through adaptive logic aligns with our goals.
2. Chaudhuri et al. (2021) compared Round Robin, Feedback (MONOPOLY), DQN, and A2C for intelligent traffic signal control using SUMO. Their findings, especially the superior performance of A2C, directly inform our selection of algorithms and evaluation metrics.
3. Mekuria & Werkhoven (2022) showed how SUMO can be used to optimize EV arrival times via traffic signal adjustments. Their work validates our use of SUMO for simulation-based testing.

Pospíchal (2021) modeled blue-light EV behavior in SUMO, including preemption and lane changes, similar to how we simulate emergency conditions in our study.

4. Suet al. (2022) used A2C for EV prioritization and achieved reduced travel times with zero collisions. This supports our decision to evaluate A2C alongside traditional methods.

Together, these works support our targeted evaluation of three control strategies—Round Robin, Feedback, and A2C—under blue-light emergency conditions using SUMO and TraCI.

Research Question

“How does the presence of blue light indicators on emergency vehicles impact the effectiveness of intelligent traffic signal control algorithms in real-world urban intersections like ABS, when simulated using SUMO and TraCI?”

Methodology

a. Simulation Tools

We used SUMO (Simulation of Urban Mobility) as the primary simulation environment. TraCI (Traffic Control Interface) was employed to interact with the simulation in real time, allowing for dynamic traffic light manipulation.

b. Data Collection

On-site vehicle counts and movement patterns for each lane manually recorded at the ABS intersection.

c. Network Design

A real-world ABS, kroanch intersection map was implemented using NetEdit and OpenStreetMap data. Multiple routes were modeled with defined emergency and normal vehicle flows shows in Fig. 1

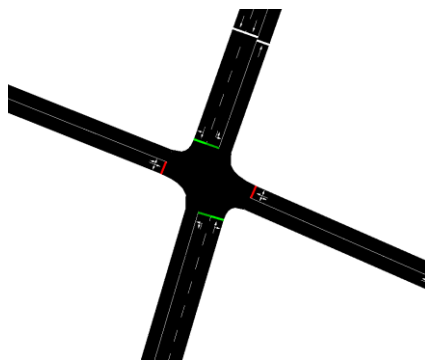


Fig. 1. ABS Intersection

d. Algorithm Implementation

- Round Robin: Equal fixed-time allocation to all signal phases in a cyclic order.
- Feedback Control: Adaptive timing based on current queue length and waiting time.

- Reinforcement Learning (A2C): Agent trained using Advantage Actor-Critic to maximize long-term traffic flow efficiency.

e. Emergency Vehicle

Emergency vehicles were introduced in the simulation with blue light enabled, granting them higher priority and allowing them to override certain traffic norms as well as overtaking other vehicles to ensure rapid passage through intersections.

f. Performance Metrics

Three primary Key Performance Indicators (KPIs) were tracked in real time:

1. Average Queue Length
2. Average Waiting Time
3. Number of Collisions

Results and Analysis

g. Baseline Simulation (without algorithms)

- Queue Length: Very high, often fluctuating between 5 and 18
- Collision count: 2
- Waiting Time: Frequently peaked above 10 seconds

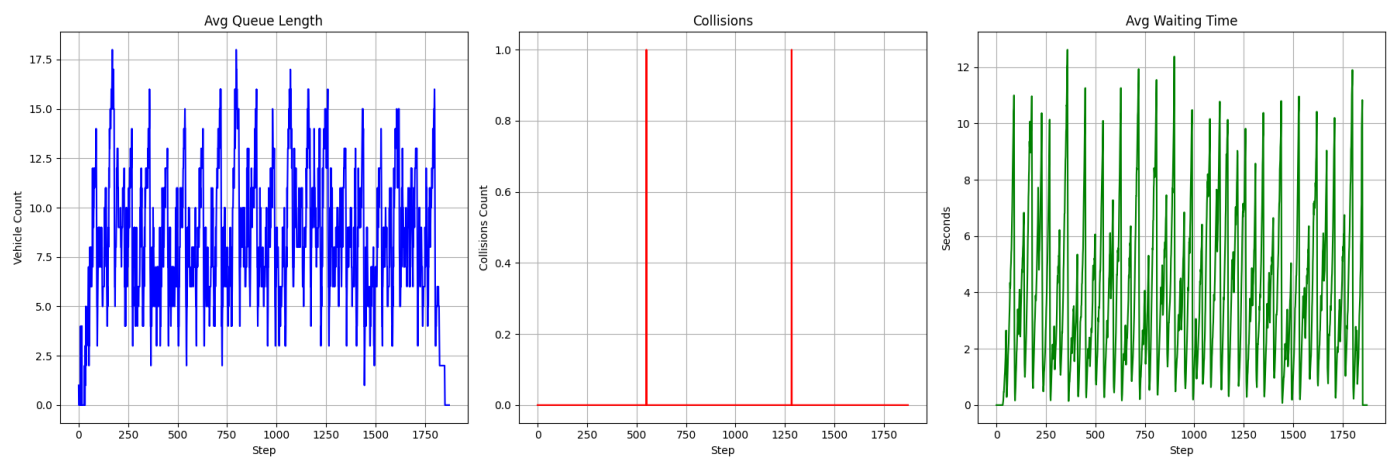


Fig. 2. Baseline

The baseline simulation without any intelligent control algorithm demonstrated significant inefficiencies in managing emergency vehicle traffic. The average queue length consistently fluctuated at high levels, often reaching above 15 vehicles, reflecting unmanaged congestion. Waiting times were also high, frequently peaking above 10 seconds, indicating delays in vehicle movement through the intersection. Additionally, two collisions occurred during the simulation, raising concerns about safety in the absence of adaptive logic shows in Fig. 2.

h. Round Robin (RR)

- Queue Length: Increased consistently
- Collision count: 8 (Maximum)
- Waiting Time: Peaked above 14 seconds

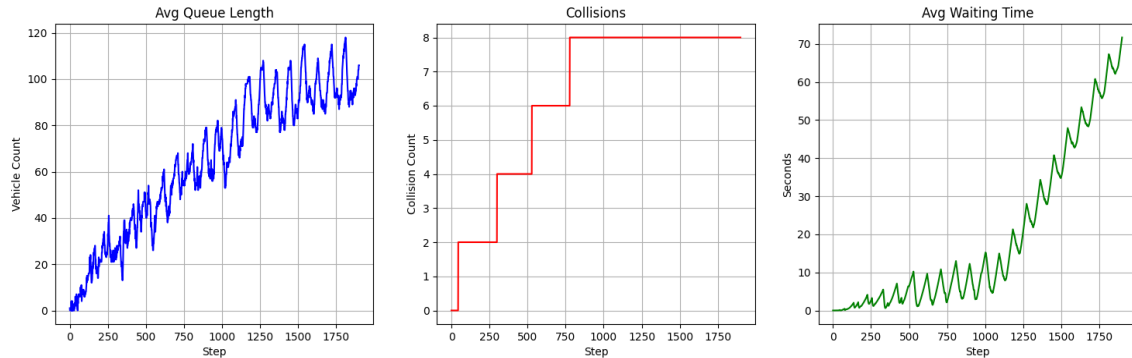


Fig. 3. Round Robin

The Round Robin approach, due to its equal and fixed timing for all directions, failed to adapt dynamically to the emergency conditions, resulting in increased congestion and a higher number of collisions. The graph reflects a sustained upward trend in queue lengths and a delayed evacuation rate, especially when blue-light vehicles are introduced as shown in Fig. 3. This underscores RR's inefficiency in emergency contexts.

i. Feedback Control

- Queue Length: Very low, mostly remained under 15
- Collision count: 0
- Waiting Time: Peaked under 1.2 seconds

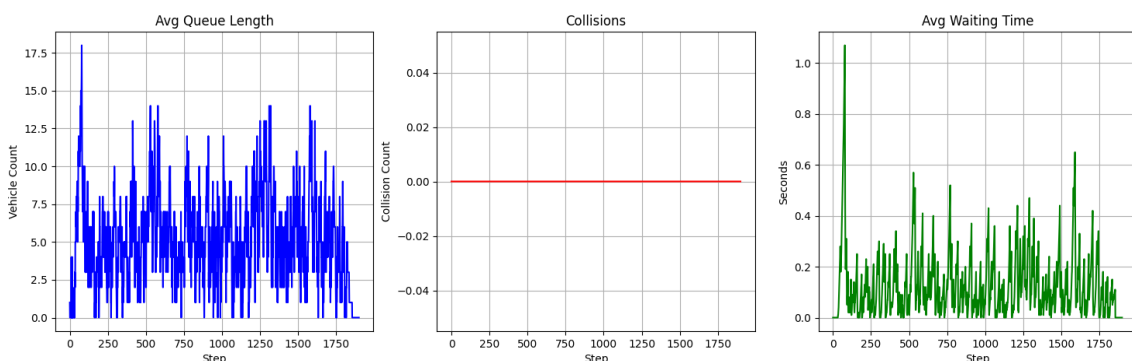


Fig. 4. Feedback Controller

The Feedback Controller showed exceptional responsiveness to real-time conditions. The graph illustrates almost negligible queue buildup and no collisions throughout the simulation. This outcome proves the effectiveness of adaptive logic in prioritizing EVs—especially under blue light conditions—allowing for quick clearance of intersections as shown in Fig. 4.

j. Reinforcement Learning (A2C)

- Queue Length: Very low, mostly remained under 7
- Collision count: 0
- Waiting Time: Peaked under 1.4 seconds

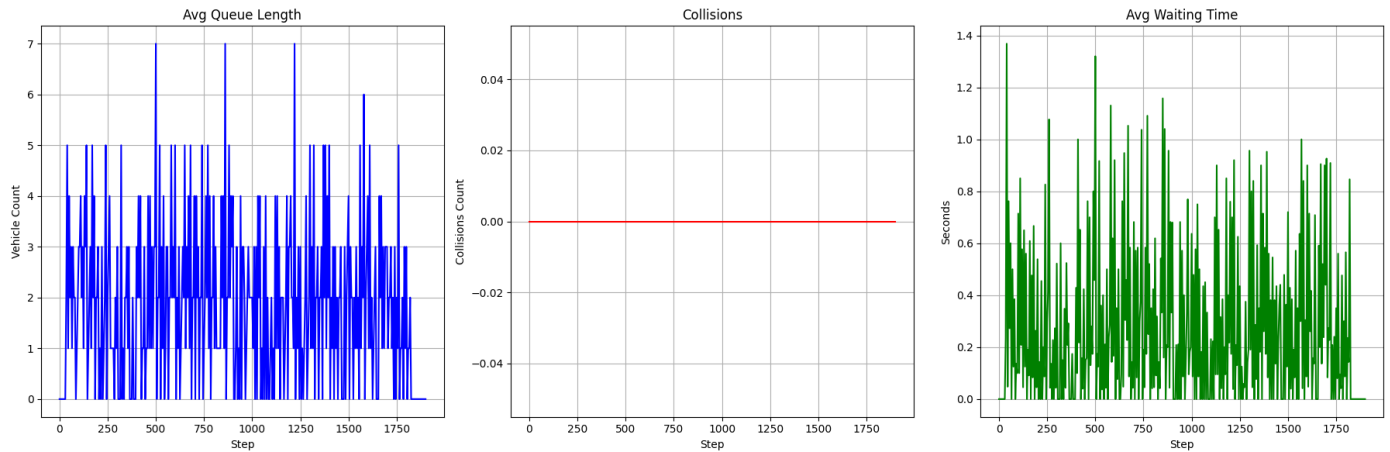


Fig. 5. RL (A2C)

The Reinforcement Learning (A2C) controller emerged as the most optimized and intelligent solution for emergency vehicle prioritization. It effectively maintained a balanced queue length—generally under 7 vehicles—while continuously adapting to dynamic traffic conditions. Remarkably, it achieved zero collisions throughout the simulation, highlighting its ability to learn safe and efficient strategies. Although brief spikes in waiting time occurred, the overall delay consistently remained below 2 seconds, reflecting superior responsiveness shown in Fig. 5. These results affirm that A2C not only ensures smooth and safe EV passage but also adapts intelligently in real time, making it the most promising control strategy among the three evaluated.

Summarised Results

A comparison of the parameters as a result of testing the different traffic light agents (RR, MONOPOLY, A2C) with different KPI are summarised in Table 1.

Control Strategy	Average Queue Length (vehicles)	Average Waiting Time (s)	Total Collisions
Round Robin	<120	<70	8
Feedback Control	~6	~0.2	0
A2C RL	~2.5	~0.5	0

Table 1.

Conclusion

This study presented a comparative evaluation of multiple Intelligent Traffic Signal Control (ITSC) strategies for emergency vehicle prioritization under blue-light conditions using SUMO and TraCI simulations. The Baseline setup, with no adaptive logic, showed moderate queue buildup and collisions, highlighting the limitations of static control in emergency scenarios.

Round Robin (RR), though commonly used for its simplicity, performed the worst across all KPIs—exhibiting the highest queue lengths, waiting times, and collisions—making it unsuitable for high-priority emergency routing. In contrast, Feedback Control demonstrated excellent performance, maintaining low queue lengths and waiting times with zero collisions, all while remaining computationally lightweight.

Among all methods, Reinforcement Learning using Advantage Actor-Critic (A2C) emerged as the most optimized and intelligent solution. It achieved the lowest average queue lengths and waiting times while ensuring complete safety, with zero collisions. A2C's adaptive decision-making and real-time learning make it highly effective for dense urban intersections where emergency vehicle response time is critical.

In conclusion, while Feedback Control offers a strong performance-to-computation trade-off, A2C is best suited for deployment in real-world high-density environments where maximizing throughput and safety for emergency vehicles is essential and computational resources are available.

References

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